sensors within a given proximity of their position. The code in listing 4.1 provides an example of this by using local coordinates to find objects within five meters of a given location using the Euclidean distance formula:

$$d\big((x_1, y_1), (x_2, y_2)\big) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This code will return the rear garden tap which is 2.6 meters from the person.

**Listing 4.1 Using spatial coordinates to perform a proximity based search query to find all sensors within five meters of a given location.**

```
my_pos = (0.1, 3) #1

sensors = [ #2
    {"id": "flow_meter_1", "name": "Rear garden tap", "pos":
    ➥(0.839, 0.5)},
    {"id": "flow_meter_2", "name": "Front garden tap", "pos":
    ➥(0.875, 18.31)},
    {"id": "water_main_meter", "name": "Main water meter", "pos":
    ➥(3.13, 21.99)},
]

dist = lambda p1, p2: ((p2[0]-p1[0])**2 + (p2[1]-p1[1])**2)**0.5 #3

print("Sensors within 5m:")
[print(f"- {s['name']}: {d:.1f}m") for s in sensors if (d := dist(my_pos,
➥s['pos'])) <= 5]
```

**#1 My position in the local coordinate system.**
**#2 Sensors, and their coordinates in the same reference system.**
**#3 Use the Euclidean distance formula to calculate the distance in meters.**

***TIP***

The ability to spatially locate sensors and the data they produce becomes significantly more important and useful the larger the area you model, and the increased density of sensors you deploy.

### 4.1.5 Derived data

Derived data results from calculations, transformations, or aggregations applied to raw data sources. This processed information transforms basic measurements into actionable insights through feature engineering, statistical analysis, and machine learning techniques.